

Instructions for this exam:

- You have 2 hours to write solutions to the four problems below. Each problem is worth 8 points.
- You must write in complete sentences and justify all of your work.
- You are NOT allowed to use outside resources, such as notes, the textbook, the internet, or people.

Exam 4, Math 3210

December 16, 2021

1. (8 pts) Using the definition of integrability (or one of the equivalent integrability criteria) prove that the function $f(x) = x$ is integrable on $[-1, 1]$ and that $\int_{-1}^1 x dx = 0$.

2. (8 pts) Prove that $\lim_{n \rightarrow \infty} \int_{1/2}^1 \cos^n(x) dx = 0$.

3. (8 pts) Consider the power series $\sum_{k=1}^{\infty} \frac{(x-5)^k}{3^k k}$.

(a) (4 pts) Find the interval of convergence I of this power series.

(b) (4 pts) If we define $f(x)$ to be the sum of the series $\sum_{k=1}^{\infty} \frac{(x-5)^k}{3^k k}$ for $x \in I$, what is $f^{(21)}(5)$?

4. (8 pts) Suppose that f has derivatives of all orders on $(-1, 1)$ and there exists a constant K such that

$$\left| f^{(n)}(x) \right| \leq n!K$$

for all $x \in (-1, 1)$, for all $n \in \mathbb{N}$. Prove that the Taylor series for f centered at 0 converges to f on $(-1, 1)$.